MATH 301: Homework 5

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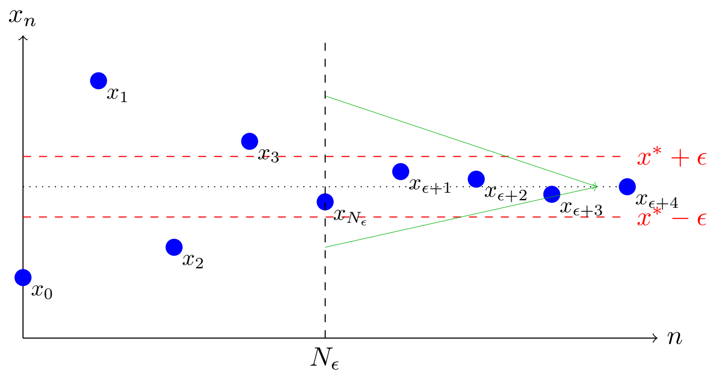
Cauchy Definition A sequence $X = (x_n)$ is called a *Cauchy sequence* if the terms of the sequence get arbitrarily close to each other as the sequence progresses. More formally,

$$\text{For every } \epsilon > 0, \text{ there exists an integer } N_\epsilon > 0, \text{ such that}$$
$$\forall\ m, n \geq N_\epsilon, \quad |x_n - x_m| < \epsilon.$$

This means that after some index N_ϵ, the distance between any two terms x_n and x_m (with n, m large enough) is less than any chosen positive number ϵ. Intuitively, the sequence becomes "clustered" together and the terms never drift far apart, no matter how far out in the sequence you go.

Example of a Cauchy Sequence



Section 3.5: 2b) Show directly from the definition that the following is a cauchy sequence:

$$x_n := \left(1 + \frac{1}{2!} + \cdots + \frac{1}{n!}\right)$$

"English": x_n is Cauchy if there's some index N_ϵ s.t. given any two indicies m, k where $N_\epsilon \leq m, k$. The differences of the two indicies is "sufficently small": $|x_m - x_k| < \epsilon$

Define terms:

$$x_m := \sum_{i=m \ | \ N_\epsilon < m}^{\infty} \frac{1}{m!} \qquad x_k := \sum_{i=k \ | \ N_\epsilon < k}^{\infty} \frac{1}{n!}$$

Goal (Show the Following):

$$\forall\ \epsilon > 0, \left| \sum_{i=m \ | \ N_\epsilon < m}^{\infty} \frac{1}{m!} - \sum_{i=k \ | \ N_\epsilon < k}^{\infty} \frac{1}{n!} \right| < \epsilon$$

$$\equiv \left|1 + \frac{1}{2!} + \frac{1}{m!} - 1 - \frac{1}{2!} - \frac{1}{n!}\right| < \epsilon$$

Suppose $n < m$:

$$\equiv \left|\frac{1}{(n+1)!} + \dots + \frac{1}{m!}\right| \leq \left|\frac{1}{2^{n+1}} + \frac{1}{2^{n+2}} + \ \dots \ + \frac{1}{2^m}\right| \equiv \sum_{k=n+1}^{\infty} \left(\frac{1}{2}\right)^k$$

$$\sum_{k=n+1}^{\infty} \left(\frac{1}{2}\right)^k = \frac{\frac{1}{2^n}}{1 - \frac{1}{2}} = \frac{1}{2^{n-1}}$$

$$-\frac{1}{n} \leq \frac{1}{2^{n-1}} \leq \frac{1}{n} \implies \lim \frac{1}{2^{n-1}} = 0$$

$$-\frac{1}{2^{n-1}} \leq |x_m - x_n| \leq \frac{1}{2^{n-1}} \implies \lim |x_m - x_n| = 0$$

$$\lim |x_m - x_n| = 0 \implies \forall\ \epsilon > 0, \exists\ N_\epsilon > 0, \ |\ \forall\ m, n \geq N_\epsilon, |x_n - x_m| < \epsilon.$$

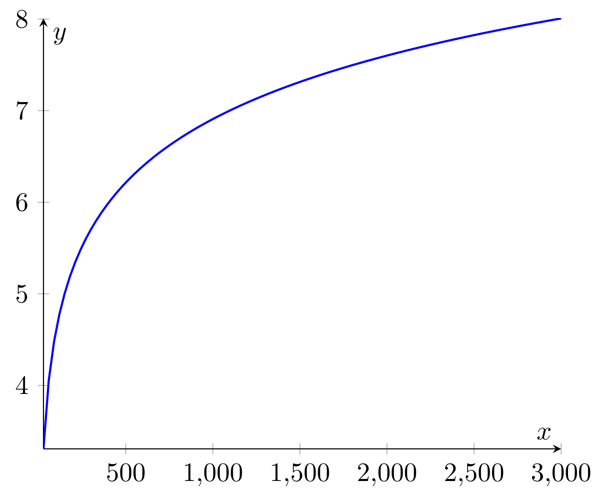
Section 3.5: 3c) Show directly from the definition that the following is NOT a cauchy sequence:

$$ln(n)$$

Negation of Cauchy

$$\exists\ \epsilon > 0\ s.t.\ \forall\ n > 0, \exists\ m, k \geq n\ s.t.\ |x_n - x_m| \geq \epsilon.$$

Graph of $ln(x)$



$$x_m := ln(2n) \qquad x_n := ln(n)$$

Consider:

$$ln(2n) - ln(n) = ln(\frac{2n}{n}) = ln(2) \approx 0.6931$$

$$\epsilon \in (0, 1] \implies \exists\ \epsilon\ s.t.\ 0.6931 > \epsilon$$

$$\therefore x_n \text{ is not Cauchy}$$

Section 3.5: 4 Show directly from the definition that if (x_n) and (y_n) are Cauchy sequences, then $(x_n + y_n)$ and $(x_n y_n)$ are Cauchy sequences

$$x_n + y_n := \{x_n + y_n\ \forall\ n\} \qquad x_n \cdot y_n := \{x_n \cdot y_n\ \forall\ n\}$$

$$|x_m - x_n| < \epsilon \implies \forall\ \epsilon > 0, \exists\ n_\epsilon\ s.t.\ \forall\ m, n > n_\epsilon, |x_m - x_n| < \epsilon$$
$$|y_m - y_n| < \epsilon \implies \forall\ \epsilon > 0, \exists\ n_\epsilon\ s.t.\ \forall\ m, n > n_\epsilon, |y_m - y_n| < \epsilon$$

$$|x_m + y_m - x_n - y_n| = |(x_m - x_n) + (y_m - y_n)| \leq |x_m - x_n| + |y_m - y_n| < 2\epsilon$$

$$|x_m - x_n| \leq |x_m| + |-x_n| \implies |y_m|\ |x_m - x_n| \leq (|x_m| + |-x_n|)\ |y_m|$$
$$|y_m - y_n| \leq |y_m| + |-y_n| \implies |x_n|\ |y_m - y_n| \leq (|y_m| + |-y_n|)\ |x_n|$$

$$|x_m y_m - x_n y_n| = |x_m y_m - \mathbf{x_m y_n} + \mathbf{x_m y_n} - x_n y_n|$$
$$= |x_m\ (y_m - y_n) + y_n\ (x_m - x_n)| \leq |x_m| \cdot |y_m - y_n| + |y_n||x_m - x_n|$$
$$|x_m| \cdot |y_m - y_n| + |y_n||x_m - x_n|$$

$$|x_m - x_n| < \epsilon \implies \exists\ M_x\ s.t.\ \forall\ n, x_m < M_x$$
$$|y_m - y_n| < \epsilon \implies \exists\ M_y\ s.t.\ \forall\ n, y_m < M_y$$

$$M_x \cdot |x_m - x_n| < \epsilon \cdot M_x$$
$$M_y \cdot |y_m - y_n| < \epsilon \cdot M_y$$

$$\implies |x_m y_m - x_n y_n| < M_x \cdot |y_m - y_n| + M_y |x_m - x_n| < 2 M_y M_y \epsilon$$
$$\frac{|y_m - y_n|}{M_y} + \frac{|x_m - x_n|}{M_x} < 2\epsilon$$
$$\frac{|y_m - y_n|}{2M_y} + \frac{|x_m - x_n|}{2M_x} < \epsilon$$

Section 3.5: 7 Let (x_n) be a Cauchy sequence such that x_n is an integer for every $n \in \mathbb{N}$. Show that (x_n) is ultimately constant.